Practical no. 7B

Aim: Create a Student Record Management PWA.

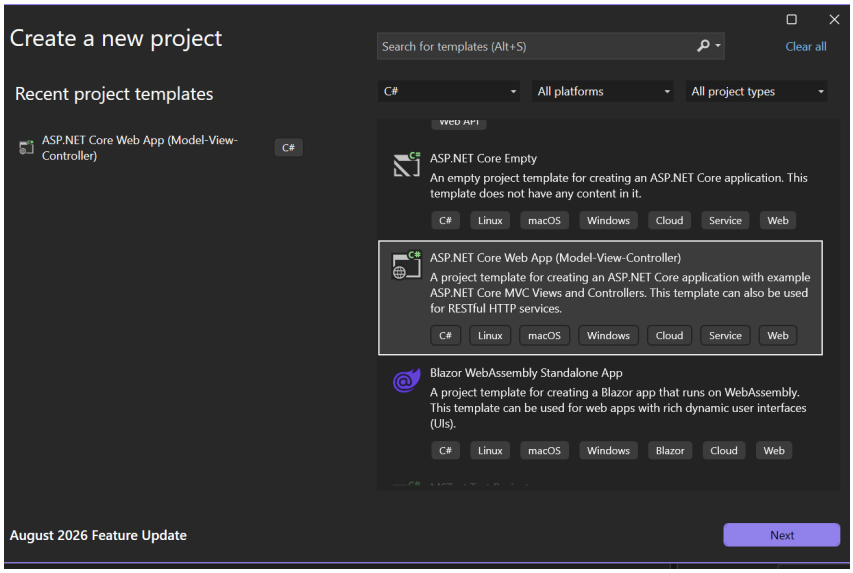
Software Required:

Visual Studio 2026

Procedure:

STEP 1: CREATE NEW PROJECT

1. Open Visual Studio 2026.
2. Select Create a new project.
3. Search for ASP.NET Core Web App (Model-View-Controller).
4. Click Next.
5. Set the Project Name to Practical No. 6b.
6. Select a suitable location and click Next.
7. Select the installed/latest .NET version and keep the default options.
8. Click Create.



STEP 2: CREATE HOME PAGE

Open Views → Home → Index.cshtml and replace its contents with:

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Records</title>

    <meta name="viewport" content="width=device-width, initial-scale=1">
```

```
    <link rel="stylesheet" href="site.css">

</head>

<body>

    <h1>Student Record Management</h1>

    <h2>Student Records</h2>

    <table>
        <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
        </tr>

        <tr>
            <td>101</td>
            <td>Rahul</td>
            <td>BSc IT</td>
            <td>85</td>
        </tr>

        <tr>
            <td>102</td>
            <td>Priya</td>
            <td>BSc IT</td>
            <td>91</td>
        </tr>

        <tr>
            <td>103</td>
            <td>Amit</td>
            <td>BSc IT</td>
            <td>78</td>
        </tr>
    </table>

    <script src="site.js"></script>

</body>
```

```
</html>
```

Step 3: Open site.css
Open:

wwwroot → css → site.css

Add the following CSS:

```
body {
    font-family: Arial;
    text-align: center;
    background-color: #f2f2f2;
}

h1 {
    background-color: blue;
    color: white;
    padding: 20px;
}

table {
    margin: auto;
    border-collapse: collapse;
    background-color: white;
}

th, td {
    border: 1px solid black;
    padding: 10px;
}

th {
    background-color: blue;
    color: white;
}
```

Step 4: Register the Service Worker
Open:

wwwroot → js → site.js

Add:

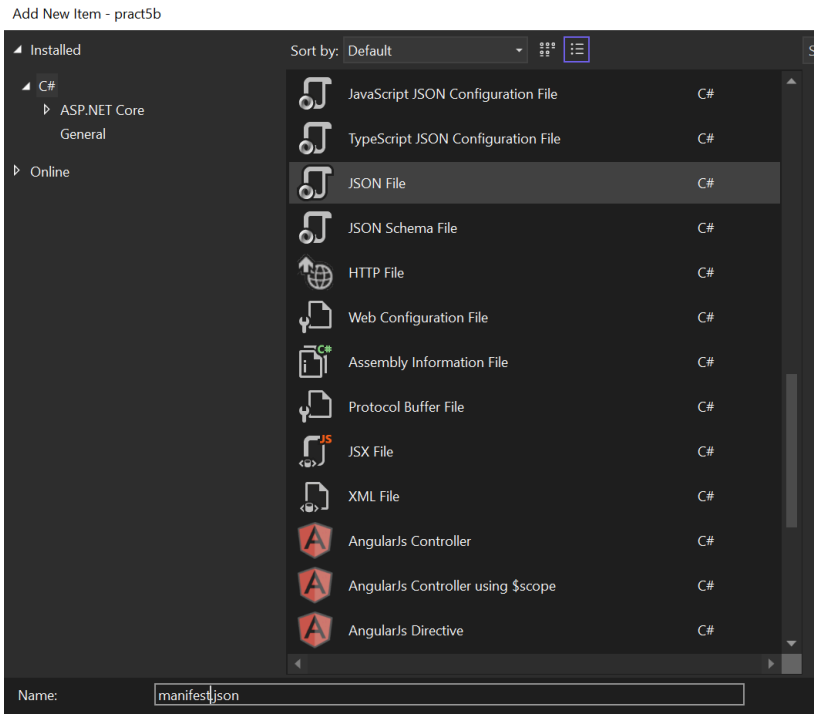
```
if ("serviceWorker" in navigator) {

    navigator.serviceWorker.register("service-worker.js")
        .then(function () {
            console.log("Service Worker Registered");
```

```
    })
    .catch(function (error) {
        console.log("Service Worker Error:", error);
    });
}
```

Step 5: Create manifest.json

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select json file.
4. Enter File Name: **manifest.json**
5. Click Add.



Enter:

```
{
 "name": "Student Record Management",
 "short_name": "Student PWA",
 "start_url": "/",
 "display": "standalone",
 "theme_color": "blue",
 "background_color": "white"
}
```

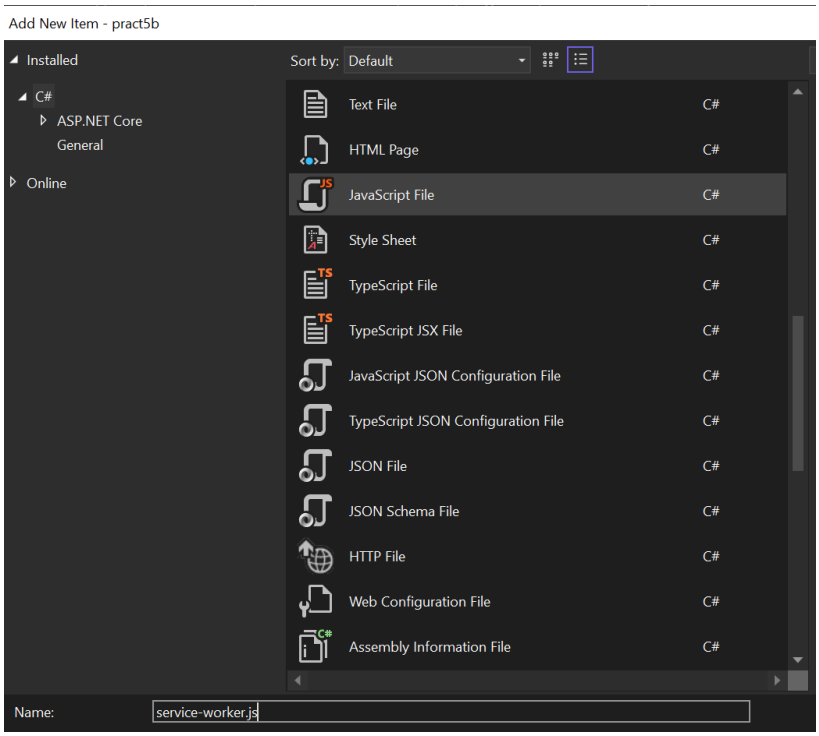
Step 6 : Register the Manifest
Open Views → Shared → _Layout.cshtml.

Inside the <head> section, add:

<link rel="manifest" href="/manifest.json">
<meta name="theme-color" content="#0d6efd">

Step 7: Create Service Worker

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select JavaScript file.
4. Enter File Name: **service-worker.js**
5. Click Add.



Write the following code in the service-worker.js file.

service-worker.js

```
const CACHE_NAME = "student-pwa-v1";
const FILES = [
    "/",
    "/Home/Index",
    "/css/site.css",
    "/js/site.js",
```

```
  "/manifest.json"
];

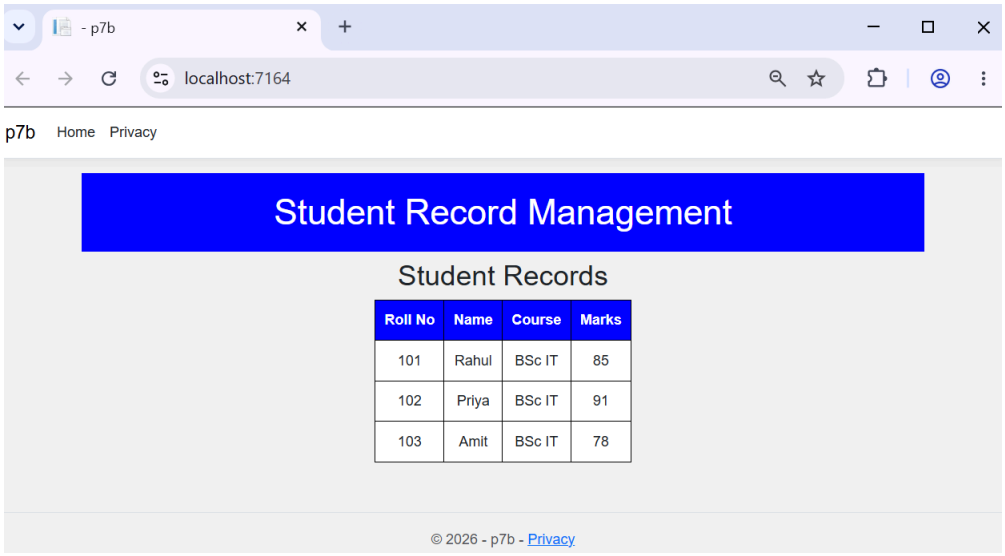
self.addEventListener("install", function (event) {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(function (cache) {
        return cache.addAll(FILES);
      })
  );

});

self.addEventListener("fetch", function (event) {
  event.respondWith(
    caches.match(event.request)
      .then(function (response) {
        return response || fetch(event.request);
      })
  );
});
```

Step 8: Run the Application

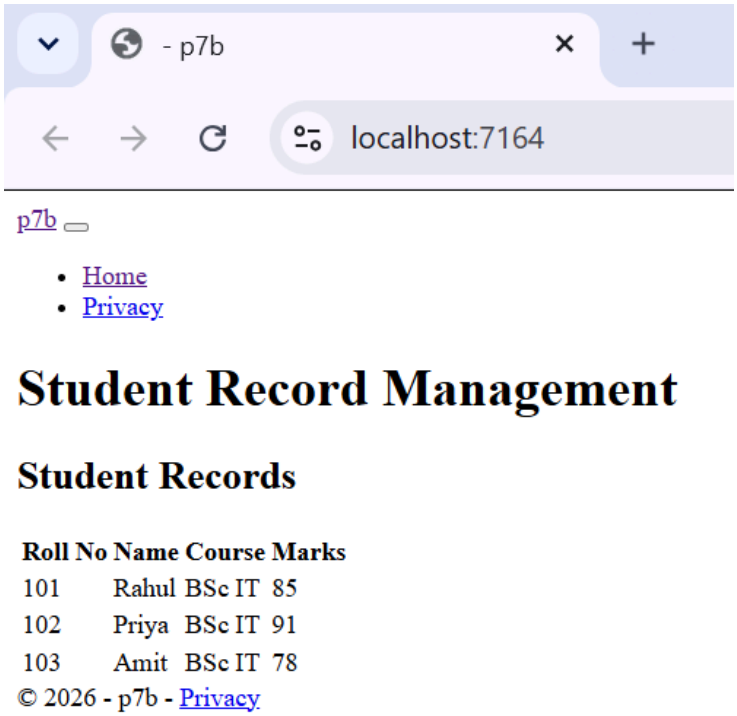
Press F5



Step 9: Test Offline Mode

1. Press: F12

2. Open: Network Tab
3. Select: Offline
4. Refresh page.



Step 10: Install the PWA

1. Click Chrome Menu: ⋮ (Three Dots)
2. Select: Cast, Save and Share
3. Then click: Install Page as App or Install College Information Portal